

Hospital Patient Management System

Course: Assembly Language Programming (Emu8086)

Project Type: Software-Based Project

Group Members: 3 Students

Project Overview

The **Hospital Patient Management System** is an Assembly Language–based software developed using **Emu8086**.

The system helps hospital staff manage patient information efficiently through a console-based interface.

The project focuses on implementing core data handling operations using **Array/Stack** and **Procedures/Macros**, fulfilling course requirements.

Technology & Tools

- Programming Language: **Assembly Language**
 - Emulator: **Emu8086**
 - Data Structure Used: **Array + Stack**
 - Programming Concept Used: **Procedures & Macros**
 - Interface: Console Based Menu System
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System Objectives

- Store patient records efficiently
- Manage hospital operations digitally
- Practice Assembly Language data handling
- Implement structured modular programming

Project Features (Minimum 6 Features)

Feature 1: Add New Patient

Assigned Member: Member 1

Description:

Allows user to add a new patient record.

Stored Information:

- Patient ID
- Name
- Age
- Gender

Implementation Concept

- Patient data stored using **ARRAY**
 - Input handled via procedure
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Feature 2: View All Patients

Assigned Member: Member 1

Description:

Displays all stored patient records sequentially.

Implementation Concept

- Array traversal
 - Loop structure
 - Display procedure
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Feature 3: Search Patient

Assigned Member: Member 2

Description:

Search patient using Patient ID.

Implementation Concept

- Linear search algorithm
 - Procedure-based search module
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Feature 4: Update Patient Information

Assigned Member: Member 2

Description:

Modify existing patient data.

Implementation Concept

- Array index update
 - Validation macro
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Feature 5: Delete Patient Record

Assigned Member: Member 3

Description:

Removes a patient from system.

Implementation Concept

- Data shifting inside array
 - Record management procedure
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Feature 6: Emergency Stack Management

Assigned Member: Member 3

Description:

Emergency patients handled using **STACK** structure.

Operations

- PUSH → Add emergency patient
- POP → Treat emergency patient

Implementation Concept

- Stack data structure
- Procedure + Macro usage